



Software Requirements Specification

Project: Rock Cannons Tracker

Authors: Joshua Allan, Dylan bardsley, Caerwyn Kerby, V Baxter, Hang Gao, Shuai Wang, Zhongyue Wang

Date: February 2026

CODE BLOODED

1 Introduction

1.1 Purpose

This document specifies the software requirements for the Rock Cannons Tracker web application. It is intended for the project team, Lecturers, and EvoMetric as the industry partner, to provide a clear, shared understanding of what the system will do and the constraints under which it will operate. The purpose of the software is to support enthusiasts, historians, hikers, and local communities in discovering, recording, and exploring historic rock cannon sites, primarily in North Wales, through a map-centric, data-driven responsive web application.

1.2 Scope

The Rock Cannons Tracker is a responsive web application that allows users to report new rock cannon sites, browse existing records, view site details, and navigate to locations via an interactive map. Core functionalities include user reporting with photos and GPS coordinates, an interactive map of sites, detailed site pages with metadata and timelines, search and filtering tools, and basic navigation/directions integration via external mapping services. The system will focus on software only (no custom hardware), and is designed for modern web browsers on desktop and mobile devices.

1.3 Definitions, Acronyms, and Abbreviations

- Rock cannon: Historic feature where holes were drilled into rock and filled with black powder to create ceremonial explosions.
- GPS: Global Positioning System, used to locate sites and users.
- UI: User Interface.
- API: Application Programming Interface used for communication between frontend and backend, and with third-party services.
- PWA: Progressive Web Application, a web app with installable and offline capabilities.
- CRUD: Create, Read, Update, Delete operations on data entities.

1.4 References

- Project brief: Rock Cannons Tracker, EvoMetric Projects (Company 2), provided as part of the university module documentation.
- Generic SRS template provided by the university ("Generic-SRS.pdf").
- Griff R. Jones, *The Rock Cannon of Gwynedd* (2002), for domain context (not a functional requirement source).

1.5 Overview

Section 2 provides an overall description of the system, including its context, key functions, user characteristics, constraints, and assumptions. Section 3 details the main system features, each with description, stimulus/response flows, and functional requirements. Section 4 describes external interface requirements, such as user interface characteristics and integration with external software and communication protocols. Section 5 outlines the system's non-functional requirements, including performance, security, and reliability expectations.

2 Overall Description

2.1 Product Perspective

The Rock Cannons Tracker is a new, standalone responsive web application deployed to the cloud, with a separate backend API and database. It operates independently but integrates with external mapping services (e.g., Google Maps, OpenStreetMap/Mapbox) for map tiles and directions, and with a cloud image storage service for managing uploaded photos. It will not interface with bespoke hardware or legacy internal systems, but may consume publicly available geographic and heritage datasets in future iterations.

2.2 Product Functions

At a high level, the system will:

- Allow users to submit new rock cannon site reports, including GPS coordinates, descriptive metadata, and photos.
- Display existing rock cannon sites on an interactive map, supporting panning, zooming, and site selection.
- Provide detailed site pages with metadata (e.g., hole count), photos, user-submitted narratives, and a timeline of reports or notable events.
- Offer search and filtering capabilities by location, date recorded, hole count, and proximity to a chosen location.
- Provide links or handoff to external mapping services for navigation and directions to a selected site.
- Optionally offer social sharing features to promote awareness of rock cannons as cultural and historical assets.

2.3 User Characteristics

- Casual visitors: Members of the public with basic web browsing skills; they primarily explore the map and view site information.
- Contributing users: Enthusiasts, hikers, and local residents comfortable with using forms, GPS/location permissions, and photo uploads; they submit new sites and contribute narratives and comments.

- Heritage professionals/researchers: Users with an interest in systematic data, who may require accurate coordinates, structured metadata, and the ability to search/filter by specific attributes.
- Administrators/Moderators: Technically competent users (project team or EvoMetric staff) who manage data quality, review and approve new submissions, and handle inappropriate content.

2.4 Constraints

- The system must run in modern web browsers (e.g., latest versions of Chrome, Firefox, Edge, Safari) on desktop and mobile devices; older browsers may not be supported.
- Development timeframe and resources are limited to the academic project schedule and the student team.
- Only widely available third-party services with free or educational tiers (e.g., free map tiles, free image hosting tiers) will be used.
- All handling of personal data and location information must align with applicable data protection regulations (e.g., GDPR) as interpreted for a student project.
- The application will be English first; additional languages may be optional future work.

2.5 Assumptions and Dependencies

- Users have internet access and can enable location services on their devices for GPS features.
- External map and image hosting services are available, stable, and within free usage limits for the duration of the project.
- The accuracy of rock cannon site information depends on user submissions and limited verification; the system does not guarantee full archaeological accuracy.
- Future integration with national heritage datasets is out of scope for the initial release but may be considered in later phases.

3 System Features

3.1 Feature 1: User Reporting of Rock Cannon Sites

3.1.1 Description and Priority

This feature allows users to submit new rock cannon site reports, including location, descriptive metadata, and photos, so that the database of sites can grow over time. It is a critical feature because user-generated content is central to the value of the application. Reports may require administrator approval before appearing publicly to ensure data quality.

3.1.2 Stimulus/Response Sequences

- A logged-in user selects “Report a site” from the interface; the system displays a form to capture GPS coordinates, hole count, description, visit date/time, and optional narrative, and allows one or more photo uploads; upon submission, the system validates the data, stores it, and marks the report as pending review.
- An administrator views a list of pending reports; upon selecting a report they can approve, edit, or reject it; if approved, the associated site becomes visible on the map and searchable to all users.
- A user attempts to submit a report without required fields (e.g., missing coordinates or description); the system displays validation messages and prevents submission until the form is corrected.

3.1.3 Functional Requirements

- FR-1: The system shall provide a “Report a site” form that captures at least coordinates, hole count, short description, visit date, and optional narrative text.
- FR-2: The system shall allow users to upload one or more photos when submitting a site report, with an upper size and file type restriction.
- FR-3: The system shall automatically associate the submitting user’s identity with each report.
- FR-4: The system shall validate form fields and display clear error messages if required information is missing or invalid.
- FR-5: The system shall store submitted reports in a persistent database with a status field (e.g., pending, approved, rejected).
- FR-6: The system shall provide an administrator interface to view, approve, edit, or reject pending reports.
- FR-7: The system shall only display approved reports as public sites on the map and in search results.

3.2 Feature 2: Interactive Rock Cannon Map

3.2.1 Description and Priority

This feature presents rock cannon sites on an interactive map that users can explore by panning, zooming, and selecting markers. It is critical, because it is the primary way users discover and navigate between sites, and it visually represents the geographic distribution of rock cannons.

3.2.2 Stimulus/Response Sequences

- A user opens the home page; the system displays a map centered on a default region (e.g., North Wales) with markers for all approved sites.
- The user zooms or pans the map; the system updates the visible markers accordingly without reloading the entire page.

- The user clicks a marker; the system displays a brief popup with the site name and key details and provides a link to the site's detailed page.
- The user enables "use my location" in the browser; the system centers the map around the user's approximate location and may highlight the nearest rock cannon sites.

3.2.3 Functional Requirements

- FR-8: The system shall display an interactive map with markers representing all approved rock cannon sites.
- FR-9: The system shall support basic map interactions, including zooming and panning.
- FR-10: The system shall display a popup with key information when a user selects a site marker.
- FR-11: The system shall provide a link from a marker popup to the corresponding detailed site page.
- FR-12: The system shall offer an option to center the map on the user's current approximate location when browser geolocation is permitted.
- FR-13: The system shall update visible sites dynamically when the user changes the map view (e.g., zoom level or region)

3.3 Feature 3: Site Detail Pages and Timeline

3.3.1 Description and Priority

This feature provides detailed pages for each rock cannon site, including metadata, photos, user narratives, comments, and a timeline of notable events or reports. It is important because it supports deeper engagement, storytelling, and historical context beyond the map view.

3.3.2 Stimulus/Response Sequences

- A user selects a site from the map or a search result; the system opens the site's detail page showing name, hole count, coordinates, description, and date of creation.
- The user scrolls down; the system shows a gallery of photos, comments, and a chronological list of reports or notable events (timeline).
- A logged-in user adds a comment or narrative to the site; the system appends this entry to the site's timeline and displays it immediately.

3.3.3 Functional Requirements

- FR-14: The system shall provide a detail page for each approved site with at least name/identifier, hole count, coordinates, and descriptive text.
- FR-15: The system shall display all approved photos associated with the site in a gallery or list.
- FR-16: The system shall provide a timeline or chronological list of reports, comments, or notable events associated with the site.

- FR-17: The system shall allow logged-in users to submit comments or narratives on a site detail page.
- FR-18: The system shall provide a mechanism for administrators to remove inappropriate comments or content.

3.4 Feature 4: Search, Filtering, and Sorting

3.4.1 Description and Priority

This feature allows users to search and filter rock cannon sites by attributes such as location, date recorded, hole count, and proximity. It is important because it supports targeted exploration and research, especially as the number of recorded sites grows.

3.4.2 Stimulus/Response Sequences

- A user enters a place name or keyword into a search field; the system returns matching sites and optionally adjusts the map view to highlight them.
- A user sets filters (e.g., minimum hole count, specific region, date range); the system updates the list of sites and the markers shown on the map.
- A user selects a sorting option (e.g., nearest, most recently recorded); the system reorders the results accordingly.

3.4.3 Functional Requirements

- FR-19: The system shall provide a text-based search for site names and descriptions.
- FR-20: The system shall provide filters by region or geographic area.
- FR-21: The system shall provide filters by hole count (e.g., minimum and/or maximum).
- FR-22: The system shall provide filters by date recorded or date of first report.
- FR-23: The system shall provide a way to search or sort sites by proximity to a user-specified location or their current location.
- FR-24: The system shall update both the list of sites and the map view in response to applied search and filter criteria.

3.5 Feature 5: Directions and Navigation Links

3.5.1 Description and Priority

This feature enables users to obtain directions to a selected rock cannon site by handing off the location to an external mapping service. It is optional but desirable, as it enhances usability for field visits without implementing full in-app navigation.

3.5.2 Stimulus/Response Sequences

- A user views a site detail page and clicks “Get directions”; the system opens the default mapping application (or a new browser tab for a web map) with the site coordinates pre-filled as the destination.

3.5.3 Functional Requirements

- FR-25: The system shall provide a “Get directions” link or button on each site detail page.
- FR-26: The system shall construct a URL or deep link that passes the site coordinates to an external mapping service (e.g., Google Maps, Apple Maps, or OpenStreetMap) for navigation.

3.6 Feature 6: Social Sharing (Optional Extension)

3.6.1 Description and Priority

This feature allows users to share site pages or discoveries on social media platforms to raise awareness of rock cannons and encourage heritage tourism. It is optional but may contribute to higher marks and stakeholder interest.

3.6.2 Stimulus/Response Sequences

- A user clicks a “Share” button on a site detail page; the system offers platform-specific share options or copies a shareable link to the clipboard.

3.6.3 Functional Requirements

- FR-27: The system shall provide a shareable URL for each site detail page.
- FR-28: The system may provide dedicated share buttons that open pre-populated posts for selected social platforms, subject to available APIs and scope.

4 External Interface Requirements

4.1 User Interfaces

The user interface shall be a responsive web UI designed to work on both desktop and mobile browsers. It will include:

- A home page with a prominent interactive map and access to search and filters.
- A clear navigation structure (e.g., top navigation bar or hamburger menu on mobile) for accessing reporting, browsing, and user account features.
- Forms with labelled fields, helpful validation messages, and accessible controls (e.g., keyboard navigation, appropriate contrast).
- Site detail pages with structured layout for metadata, photos, comments, and timelines.

4.2 Hardware Interfaces

The system will not interface with bespoke hardware. It will rely on:

- Standard device sensors exposed through the browser, such as GPS (via the Geolocation API) and device camera integration for photo uploads, where supported.

4.3 Software Interfaces

The system will interface with:

- A relational database management system for persistent data storage.
- External map tile and geospatial services (e.g., OpenStreetMap, Mapbox, or Google Maps) for map rendering and directions links.
- A cloud image hosting service (e.g., Cloudinary or similar) for storing and delivering uploaded photos.
- Browser APIs for geolocation and file input.

4.4 Communication Interfaces

The web application frontend and backend will communicate over HTTPS using RESTful HTTP APIs with JSON payloads. The system will also communicate with external services using HTTPS, following their published API specifications where applicable. All communication shall be encrypted in transit.

5 System Non-Functional Requirements

5.1 Performance Requirements

- The system should load the main map view within a few seconds under normal network conditions and initial dataset sizes.
- The system should return search and filter results for typical queries with a small number of seconds of latency for the expected dataset size during the project.
- The system should support a small concurrent user base typical for a student project and demonstration environment without noticeable degradation.

5.2 Security Requirements

- The system shall require authentication for actions such as submitting reports, commenting, and accessing administrative functions.
- The system shall transmit sensitive data (e.g., login credentials) only over HTTPS.
- The system shall implement basic input validation and sanitization to reduce common risks such as SQL injection and cross-site scripting in the project's context.
- Administrative functions (e.g., approving reports, deleting content) shall only be accessible to users with appropriate roles.

5.3 Reliability Requirements

- The system should be available for demonstration and evaluation during agreed periods, with hosting configured to minimize unplanned downtime within the constraints of free or educational hosting services.
- In the event of service interruption, the system should recover when hosting is restored without data loss, assuming regular database backups or exports are performed in line with the project plan.
- The system shall handle failures of external services (e.g., map tiles not loading) gracefully by displaying informative messages or fallbacks rather than critical errors.